

Tactile Data Comics: A Step-by-step Multimodal Presentation Method on a Refreshable Tactile Display for Blind and Visually Impaired Individuals

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Abstract

Refreshable tactile displays (RTDs) could present dynamic tactile graphics to support blind and low-vision individuals in learning. However, existing research offers limited evidence on the effectiveness of dynamic tactile graphics on RTDs in supporting blind students' learning outcomes. To study the value of presenting dynamic contents in the learning context, we designed *tactile data comics*, a method that combines step-by-step presentation of tactile graphics with verbal narration. We conducted a user study with 16 visually impaired students to compare tactile data comics against verbal-only and static tactile graphic presentation approaches. Our findings show that tactile data comics significantly improves participants' content comprehension. The results highlight the potential of tactile data comics to benefit the visually impaired community, particularly in educational applications.

CCS Concepts

• Human-centered computing → Accessibility design and evaluation methods.

Keywords

Accessibility, Human-Computer Interaction, Refreshable tactile display

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1 Introduction

For centuries, tactile graphics have been used by individuals who are blind or have low vision [18]. Traditionally, these graphics have been produced on paper using methods such as vacuum forming, braille embossing, or microcapsule paper [5, 25]. However, among the tactile graphic media, refreshable tactile displays (RTDs) are unique in their ability to present dynamic content [25]. The working mechanism of RTDs is based on an array of thousands of small tactile pins, each capable of independent vertical movement to create a dynamic tactile surface. With the rise of maker technologies, various tactile display devices have emerged in the market or are currently in the prototype phase. Currently, the mainstream tactile displays on the market include Dot Pad [10], Metec.AG HyperBraille F [17], APH Monarch [7], and Orbit Graphiti [28]. The RTDs' capability in presenting dynamic contents offers promising educational potential as it could present complex concepts that are hard to communicate in static media.

Research on the application of RTDs has primarily focused on displaying static images in fields such as art [8], book diagrams, and illustrations [11, 20, 22, 24]. Researchers utilize the combination of haptic technology and RTDs to help visually impaired students better understand and access static images in textbooks, such as charts and illustrations [11, 24]. Other studies have explored the dynamic presentation of content, for example, sports matches [12, 21], UML diagrams on RTD [14], and animation segments like bird flight simulations [9]. Additionally, researchers have investigated methods to utilize refreshable tactile displays for teaching spatial information

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in maps [13, 29, 33, 34]. These applications demonstrate the potential of RTDs in enhancing the experience for the blind community, particularly in interactive and dynamic learning contexts.

However, evidence for the effectiveness of dynamic content presentation on RTDs in supporting blind students' learning remains limited. While past studies reported positive user feedback regarding learning experiences with RTDs, they relied primarily on self-reported qualitative assessments rather than controlled experiments measuring comprehension, potentially introducing bias and limiting conclusions about their educational impact [9]. Moreover, many existing studies predominantly involve older participants who are not students, leading to a gap in understanding how RTDs can effectively support blind students in educational settings [9, 27]. The lack of quantitative data in supporting (or refuting) the RTDs' pedagogical value and demographic discrepancy underscores the need for targeted research that includes blind students to accurately assess the educational potential of RTDs.

In this paper, we introduce a new dynamic presentation method—*tactile data comics*—and conduct a controlled experiment to evaluate how RTDs and dynamic presentations of tactile graphics affect blind students' comprehension of educational materials. Tactile data comics incrementally reveal complex visual information in smaller, sequential components, with each component building upon the previous one, retaining earlier content while introducing new elements. Each component is accompanied by a verbal narration that maintains cross-modal consistency to help blind users mentally construct the presented graphic.

Using this method, we conducted a controlled experiment with 16 blind students to evaluate the effectiveness of RTDs and tactile data comics in education. The technique was applied to educational materials across four subjects—mathematics, physics, maps, and literature. The study included three conditions: *verbal-only*, *verbal with static graphics*, and *tactile data comics*. After engaging with the materials, participants answered quiz-like questions related to the content, allowing us to test the effectiveness of these methods through objective quantitative data rather than relying solely on subjective learning experiences.

Our results demonstrated that tactile data comics significantly improved learning comprehension. The answer accuracy rates were 70.1% for *verbal-only*, 70.1% for *static graphics*, and 76.6% for *tactile data comics*. The effectiveness varied depending on the type of educational material, with response accuracy being particularly higher for map comprehension and understanding literature narratives when using tactile data comics. Additionally, subjective assessments indicated that participants found tactile data comics more engaging than both the *verbal-only* and *static graphics* conditions.

In summary, the contributions of this paper include: (i) the design of tactile data comics, a procedural presentation method for tactile graphics and verbal narratives that provides cross-modal consistency, and (ii) the evaluation of tactile data comics by providing quantitative experimental evidence of their effectiveness in the education field.

2 Tactile Data Comics

In this section, we describe the refreshable tactile display used to present tactile data comics, introduce the concept of tactile data

comics, and explain how they are composed and applied in our study.

2.1 Refreshable Tactile Display

In this study, we utilized an internally developed tactile display prototype to present tactile graphics (Figure 1). The device features a 15 cm by 15 cm display area and a 60×60 array of 3,600 refreshable pins (while commercially available tactile displays typically have 2,400–3,840 pins). The pins are spaced 2.5 mm apart, consistent with mainstream tactile displays on the market, and each pin is 1 mm in height. It can refresh an image in less than 1 second, enabling the presentation of dynamic content. Thus, the device shares similar features with other commercially available tactile displays.

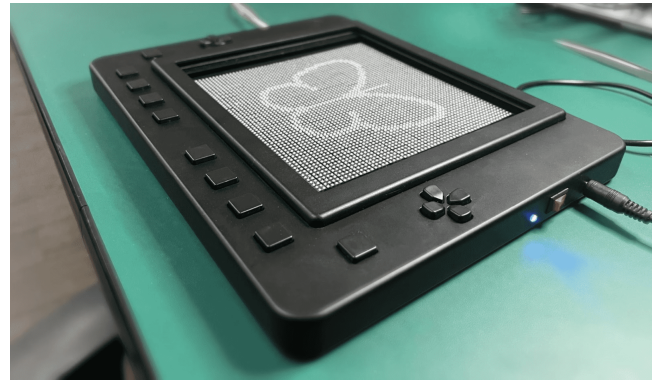


Figure 1: Internally developed tactile display showcasing a butterfly.

2.2 The Concept of Tactile Data Comics

Using the refreshable tactile display as the output device, we designed tactile data comics. Tactile data comics use a step-by-step approach that breaks down complex images into smaller, digestible components. In this method, each component builds upon the previous one, retaining earlier content while introducing new elements (Figure 2). Additionally, each component is accompanied by verbal description of the presented content. The method ensures consistency between the two modalities, haptic and auditory, to facilitate the user's comprehension of the presented material.

Tactile data comics separate themselves from the existing presentation methods for educational tactile graphics in two ways. The first feature is a discrete frame-by-frame presentation of progressively evolving content, just like comics (Figure 2). This is in contrast to a continuous presentation, like animations, in [9, 32]. We made this design choice to reduce learning difficulties when perceiving the sequence of graphics displayed on the RTDs. Its progressive nature allows users to easily identify changes between frames; each component is built upon the previous one and retains the previously presented components until it transitions to the next image.

The second feature is pairing of each component with a verbal description, aiming to help users logically piece together the parts and gain a better understanding of the graphics being presented.

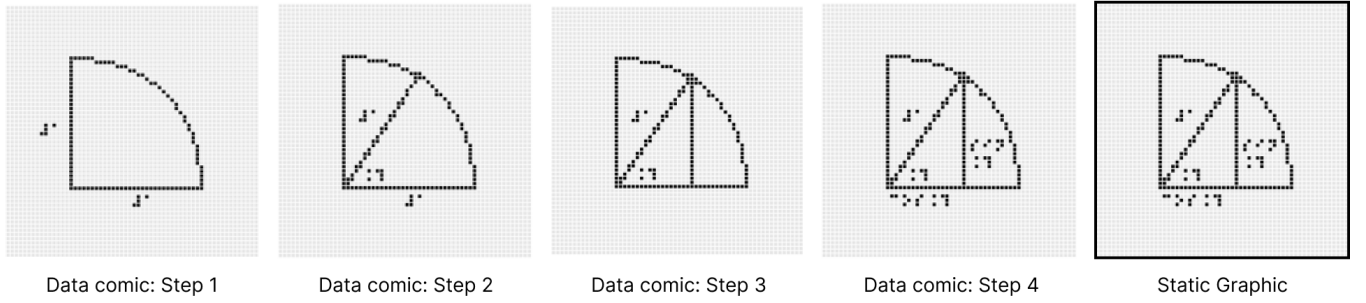


Figure 2: The four images on the left are tactile data comics of a geometric diagram explaining the Pythagorean theorem’s principles, while the outlined image on the right is a static graphic of the same principles.

The design of tactile data comics aims to extend the traditional learning experience, where verbal narration from teachers and static tactile images in textbooks were used to convey content. In the traditional method, regardless of which step of the content the teacher is introducing, the static tactile images from the textbooks remain unchanged and do not adapt to the teacher’s narration. RTDs allow us to dynamically present a graphic that matches what is being explained. Our approach enables each step of the narration to correspond to a specific part of the tactile graphic, ensuring alignment between verbal and graphical information (Figure 2). For instance, in Fig. 2, the narration begins with “*This is a quarter circle...*” and is paired with a tactile graphic in step 1 that depicts the quarter circle. As the explanation progresses, additional elements are introduced in subsequent steps, building on the prior frame. This approach ensures that blind students only interact with content directly related to the narration at each step. We believe this step-by-step, multi-modal presentation method significantly enhances information comprehension, making tactile data comics a valuable tool in blind education.

2.3 Tactile Data Comics Production

We compose tactile data comics contents by (i) manually segmenting static tactile images into step-by-step contents, and (ii) synchronizing the accompanying verbal narration to the segmented tactile contents. We break down the static tactile images based on the logical structure of the content, ensuring the completeness of each component. Then, we create a verbal description to explain the content of each component. Additionally, we focus on the overall coherence and logic of the descriptions, so that they can form a complete verbal narration.

3 Experiment

3.1 Experiment Materials

We conducted in-depth discussions with teachers from blind schools to gather the teaching materials they currently use and identify which materials could benefit from the integration of illustrations for enhanced learning. Based on this collaboration, we transformed existing illustrated materials from textbooks or created new ones, closely aligned with the curriculum, into tactile data comics. We selected four subjects—mathematics, physics, maps, and literature.

For each subject, we developed three sets of materials, resulting in a total of 12 sets of tactile comics (Table 1).

Each material was paired with step-by-step a verbal description, generated by a text-to-speech tool (<http://ttsqpt.cn>) to minimize variations in tone and reading speed between each experiment session. For mathematics, physics, and maps, the static graphic and the final data comic segment were equivalent. For literature, which involved longer passages, each material contained 1-4 static graphics, also segmented for the tactile data comics (Figure 3). The full narrations and tactile comics for the 12 materials are available in the supplemental materials.

3.2 Study Design

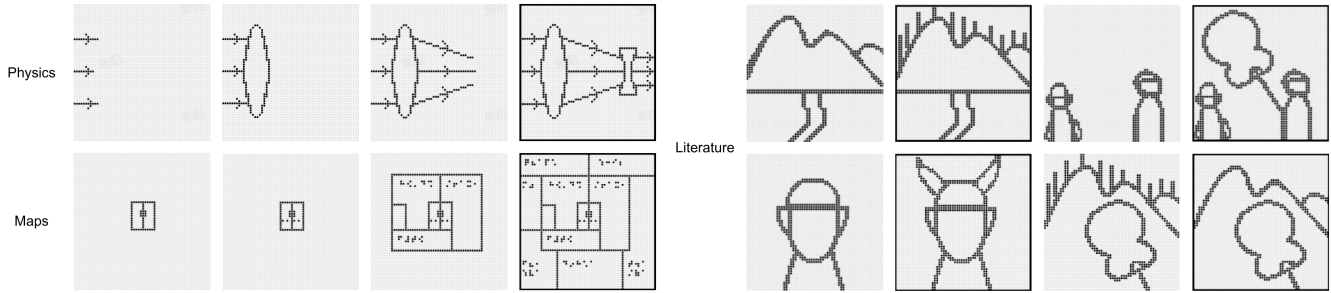
To evaluate the effectiveness of tactile data comics in conveying information and enhancing engagement for blind individuals, we conducted a within-subjects controlled study. The study tested three presentation methods: (1) verbal narration only (*verbal*); (2) verbal narration with static tactile graphics (*static*); and (3) tactile data comics (*comic*). The verbal and static conditions reflect how educational content is typically presented in classrooms, while the comic condition introduces a step-by-step approach designed to simplify complex concepts and graphics. We hypothesized that the comic condition would be the most effective in supporting comprehension.

Each participant engaged with 12 learning materials, using one presentation method per material. The order of presentation methods and learning materials was counterbalanced to prevent order effects. Participants explored the subjects in the fixed sequence of mathematics, physics, maps, and literature. During the study, the tactile display was connected to a smartphone via a Type-C cable for control. To ensure clarity, participants were notified whenever the graphic changed. To prevent variations in verbal narration from influencing the study results, we kept the audio content consistent across all three conditions. In the verbal-only and static conditions, the narration played alone. In the comics conditions, the narration synchronized with the tactile graphics. A 5-second pause was included at each section break within a material across all conditions.

After each material, participants were asked to answer six questions to assess their comprehension [15, 23]. This study’s use of question-answering provides a more direct and accurate way to verify the practical application of tactile data comics in blind education. We designed all the questions to be answerable solely through

Table 1: Themes for the twelve educational materials used in our user study. We had four subject categories: *mathematics*, *physics*, *maps*, and *literature*. For each subject, we prepared three materials.

Category	Material 1	Material 2	Material 3
Mathematics	Pythagorean Theorem	Trigonometric functions	Pythagorean identity
Physics	Reflection of light	Refraction of light	Principles of the telescope
Maps	District map	A suburban district map	A park map
Literature	“Journey to the West” Story 1	“Journey to the West” Story 2	“Journey to the West” Story 3

**Figure 3: Examples of tactile data comics for *physics*, *maps*, and *literature* among the twelve educational materials that we used for the user study. A graphic with dark border represents the static tactile graphic that we used for each material. Because the narratives for the *literature* subject were longer, the accompanying tactile data comics were longer than those for other subjects, and we had more than one static graphics per material.**

verbal narratives, without relying on tactile graphics. Half of the questions focused on recalling details from the narrations [16], while the other half were more conceptual and required participants’ overall understanding of the material, evaluating not just memory but comprehension of the information as a whole [23].

At the end of the study, we conducted UEQ-S surveys and interviews to evaluate participants’ perceptions of the different presentation methods. We used UEQ-S questions to assess each presentation method’s perceived pragmatic and hedonic qualities [30]. The UEQ-S includes seven items rated on a 7-point scale (e.g., -3 for “very obstructive” to +3 for “very supportive”) and evaluates both usability and engagement. UEQ-S’s design to assess the level of engagement and its brevity made it suitable for our study [30].

3.3 Participants

We recruited N=16 participants from a blind college in the first author’s city. The criteria for participant selection were that they must use braille. Some participants had experience with tactile displays, but none owned or used these devices in their daily lives. All participants had a basic understanding of the relevant subjects, though they were not familiar with the specific material used in this study. Their demographic data is shown in Table 2.

3.4 Procedure

Each study session took place in a classroom at the school for blind students. A research team member led each session. Sessions were audio and video recorded using two smartphones: one placed in front to capture the entire experimenter-participant interaction and another positioned above the tactile display to capture hand interactions. The session involved four stages (Figure 4):

Table 2: Demographic data of our participants. All used braille and screenreader for learning. For the onset of visual impairments, early’ refers to acquiring blindness in early childhood at the age of six or younger, and late’ represents school age between 6 and 18 (the number in brackets represents the age of visual impairment onset).

Participant	Sex	Age	Onset of blindness
1	Male	24	Late (8)
2	Female	25	Late (13)
3	Female	19	Early
4	Female	26	Late (11)
5	Female	26	Early
6	Male	23	Early
7	Female	22	Early
8	Female	22	Late (8)
9	Female	24	Late (15)
10	Male	20	Late (18)
11	Male	24	Late (7)
12	Male	25	Late (12)
13	Male	27	Late (7)
14	Female	20	Early
15	Male	20	Early
16	Male	21	Early

- (1) **Introduction:** After introducing and reviewing the study design with a participant, we obtained informed consent and collected demographic information. We explained that

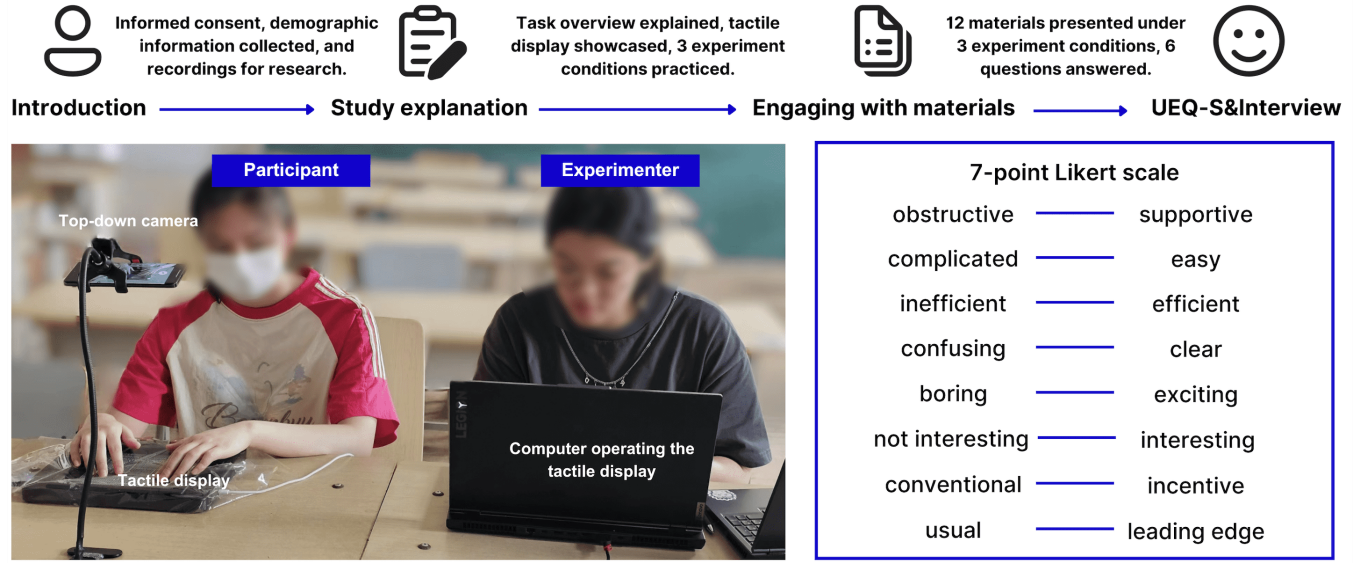


Figure 4: Overview of the study procedure, comprised of four stages: introduction; study explanation and practice; engaging with learning materials; and UEQ-S surveys and interviews.

- recordings would be used solely for academic research purposes and, upon consent, started audio and video recordings.
- (2) **Study Explanation and Practice:** The experimenter explained the task overview and learning materials. We demonstrated tactile display usage, and participants briefly experienced static and comic presentation methods.
 - (3) **Engaging with Learning Materials:** We guided participants to engage with twelve learning materials using different presentation methods, following the exposure order described above. After each material, participants answered six questions, and the experimenter recorded the correctness of the response.
 - (4) **UEQ-S& Interview:** After engaging with all materials, the experimenter used UEQ-S to gather participants' ratings of the three presentation methods. At the end of the study session, we conducted a semi-structured interview.

Each participant spent approximately 1 to 1.5 hours in completing the study session. We compensated the participants with about US\$20 for their participation.

4 Results

We report on two analyses: (i) Comprehension accuracy for three presentation methods; (ii) UEQ-S based analysis on subjective ranking of preferred presentation modes.

4.1 Comprehension Accuracy

Here, we examine how presentation methods and random effects affected comprehension of learning materials. Correctness for each question after engaging with a learning material was measured with a binary outcome (*i.e.*, correct or incorrect). Therefore, statistical tools like repeated measures ANOVA, which expects a continuous response, are not ideal. Thus, we used a generalized linear mixed

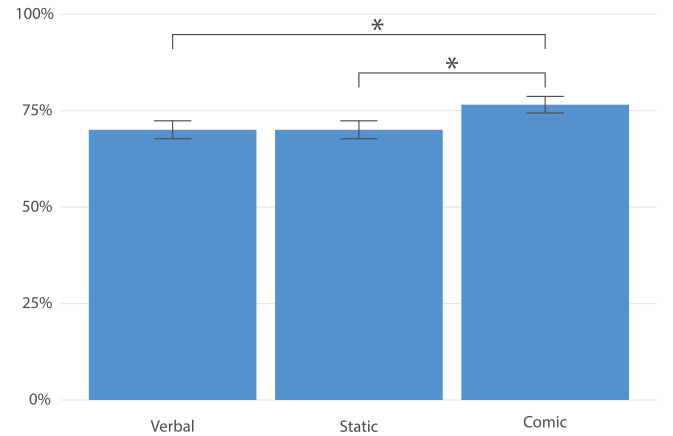


Figure 5: Comprehension accuracy for three conditions: verbal, static, and comic. The accuracies are 70.1% for verbal narration (verbal), 70.1% for verbal narration with static tactile graphics (static), and 76.6% for verbal narration with tactile data comics (comic).

model with a logit link function. The presentation method was the main effect. We treated materials' subjects and participants' onset as random effects.

Our participants answered the questions more accurately with the tactile data comics presentation. The accuracies were 70.1% for *verbal*, 70.1% for *static*, and 76.6% for *comic* (Figure 5). The difference in accuracy between *verbal* and *comic* conditions was statistically significant ($p = 0.039 < 0.05$), and so was the difference between the *static* and *comic* conditions ($p = 0.039 < 0.05$). We did not observe a statistically significant difference in accuracy

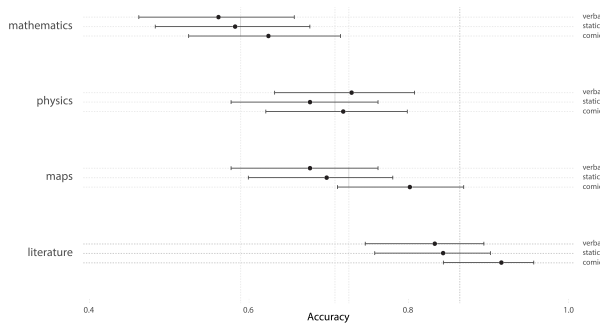


Figure 6: Comprehension accuracy for three conditions across four subjects. Each horizontal black segment shows the confidence interval of accuracy. Gray vertical bars represent per-subject mean accuracies (math: 0.59, physics: 0.71, maps: 0.73, literature: 0.87).

between the *verbal* and *static* conditions. The result supports that data comics have a positive effect in aiding comprehension, helping users better understand the learning materials.

But there seemed to be more nuance to the result; the questions' subject influenced response accuracy. Overall, participants answered literature questions most accurately, with an average accuracy of 86.5%. For map and physics questions, accuracies were 72.6% and 70.8%, respectively. Participants' accuracy in answering math questions was lowest at 59.0% (Figure 6). This result indicates the variation in participants' aptitude for answering different question types in addition to the main effect of the presentation method.

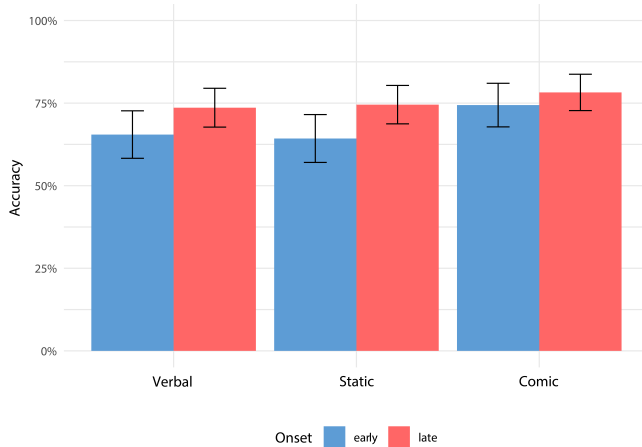


Figure 7: Comprehension accuracy: blindness onset (early vs late)

We grouped data by condition and blindness onset (early vs late) to explore possible differences in response accuracy. Figure 7 suggests that the early onset group tended to have lower accuracy scores compared to the late group. However, this accuracy difference decreased in the comic condition. This may suggest that those

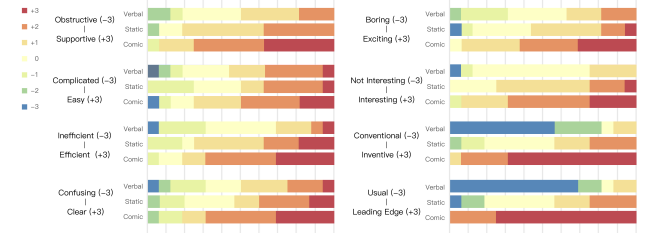


Figure 8: Distributions of UEQ scores across different interface conditions. Orange to Red segments represent positive sentiment toward the usability of the presentation method, whereas Blue to Green segments represent negative.

who acquired blindness congenitally or early in life benefited more from tactile data comics. Nevertheless, the overlapping confidence intervals suggest no significant differences and prevent us from making solid conclusions.

4.2 UEQ-S Responses

Perceived ease of interaction and engagement with the technology are important factors, particularly in a context like an educational setting, where such factors could impact retention. We collected subjective assessments of pragmatic and hedonic qualities for each presentation method. The stacked bar charts in Figure 8 correspond to distributions of the UEQ-S responses for each item. Positive sentiment, represented by '+1' to '+3', was most prevalent across all items for tactile data comics. The results show that participants felt learning with tactile data comics was easier and more engaging compared to learning with static tactile graphics or verbal narrations alone.

5 Discussion

We designed tactile data comics, a step-by-step presentation of tactile graphics enabled by refreshable tactile displays. Our goal was to improve blind users' comprehension of educational materials by procedural presentation of tactile graphics together with accompanying verbal narrations. Our study with 16 blind participants showed significantly increased response accuracy in comprehension tasks when using tactile data comics, particularly for map content comprehension and literature narrative understanding. However, we observed no evident benefit for math and physics tasks. The results highlight the potential of refreshable tactile displays' dynamic nature in designing novel interaction methods for blind users.

Tactile data comics not only enhance learning outcomes but also improve the overall learning experience by making it more interesting and engaging. Compared to verbal narration and static graphics, our participants felt that tactile data comics make the learning material more vivid and immersive, as suggested by our summary of UEQ scores. The constantly changing graphic information seemed to have improved the user's focus and attention during the learning process. In contrast, relying solely on verbal information for extended periods demands sustained attention, making it harder to maintain concentration. After the study one of the participants said, "*The tactile presentation method helps me maintain*

focus because the graphics are changing. Each change in the graphic brings new information, ensuring high levels of engagement” (P8). In the future work, we will analyze the interview responses in more detail to accrue more qualitative insights.

A significant challenge in increasing tactile display adoption and developing applications is content authoring. Our process of creating static and comic representations from existing materials was manual and time-consuming. How can we more efficiently translate visual content into tactile data comics, particularly given their step-by-step structure and pictorial nature? While past work has explored authoring methods for tactile displays [2–4, 6, 26, 31], creating good tactile graphics remains challenging [25]. Given the rise of generative AI, an exciting next step would be exploring whether image generation techniques can automatically create high-quality tactile content. Mukhiddinov *et al.* have initial work in this direction[19], but more research is needed. to effectively generate good tactile graphics at scale.

In this study, we conducted a quantitative analysis of comprehension accuracy and UEQ-S results. In the future, we plan to analyze qualitative interview responses to provide a more comprehensive evaluation and validation of tactile data comics. Additionally, this work focused on college students, but we recognize that children of different age groups have varying abilities to comprehend shape information through touch [1]. To gain a more concrete understanding of the effectiveness of different tactile representations for learning, a larger-scale investigation is needed.

6 Conclusion

In this work, we designed and evaluated tactile data comics, a method for presenting tactile graphics in a step-by-step manner to take advantage of the refreshable nature of the tactile display. Through a user study with 16 blind students, we demonstrated that tactile data comics facilitate comprehension of educational materials compared to control conditions (verbal-only presentation and static tactile graphic presentation). The experimental results, including quantitative data collected through a question-and-answer format and qualitative data from the UEQ-S questionnaire, provide robust and direct evidence for the effectiveness of RTDs in educational applications. These findings suggest that RTDs and tactile data comics offer a promising approach for enhancing the accessibility and effectiveness of tactile learning tools, and warrant further exploration in educational contexts.

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